Haniel Thomson

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Summary

Software Engineer building scalable and data-intensive AI systems. Built low latency backend infrastructure at Amazon Robotics supporting the Package Identification Technology Amazon unveiled at re:MARS 2022, the AI scanning systems that are now part of the largest warehouse robot fleet on earth, more than a million robots in 300+ facilities. Currently building LLM agent pipelines automating manufacturing operations and RFQ processing for industrial customers.

Experience

Software Engineer **Sep 2025 – Present**
Tarvo AI *Remote, USA*

- Serving as a founding engineer on a small team building an AI driven web platform that automates manufacturing quote generation from inbound RFQ emails, owning both the agent pipeline design and the customer facing pricing tooling.
- Created multi-step LLM agent pipelines using LangGraph to classify inbound RFQs and extract structured fields, achieving 98% extraction accuracy and cutting manual quote-processing time by 75% across 500+ RFQs/week.
- Built a three-layer evaluation harness spanning unit, integration, and shadow evals with LLM-as-judge scoring to validate agent output quality before production rollout.
- Integrated Inngest as a durable execution layer around the pipeline for back pressure and fault tolerance, and connected the Microsoft Graph API and Extend.ai for email ingestion and document-to-text extraction.
- Designed a deterministic pricing engine using topological sort graph execution that lets customers define their own pricing logic visually in React Flow, with the LLM limited to populating variables for fully auditable output.
- Built shortcut-based navigation patterns, improving interaction speed and reducing reliance on mouse-driven workflows.
- Created serverless REST APIs with AWS Lambda and SST and instrumented them with Sentry, lowering infrastructure usage and improving production observability.

Software Engineer (Contract) **Dec 2024 – Aug 2025**
Amazon Robotics *Westborough, MA*

- Contracted to support hardware-in-the-loop testing initiatives for Amazon Robotics' Scanless barcode recognition systems, working across release infrastructure, test automation, and system integration for customer Lumos.
- Architected and deployed fully automated release infrastructure using AWS CDK and CloudFormation, slashing deployment latency by 90% and eliminating manual intervention for artifact delivery to S3.
- Built a low-latency, LMAX Disruptor based test harness that collected and validated data from multiple sources, supporting broader hardware-in-the-loop initiatives.
- Developed a sensor simulation harness using Kotlin and Docker to automate barcode recognition testing, removing manual QA bottlenecks and recovering 30+ engineering hours monthly.
- Integrated the team's metrics service with a higher-level metrics service in C++ and wrote gMock test coverage to validate the integration.
- Set up AWS SNS to publish barcode decode results to customer Lumos, enabling Amazon's systems to communicate directly with the customer's infrastructure.
- Optimized edge fleet provisioning by migrating deployments to geo-proximate AWS regions and ECR, reducing device setup time by 60% and enabling rapid fleet scaling closer to the customer.
- Benchmarked system latency by designing bash-based workload simulations, collecting 160+ hours of performance data to identify and resolve critical bottlenecks and facilitate test automation.
- Increased test coverage by 10% across 10+ Python and Java packages using pytest and JUnit.
- Traced a release-blocking production incident to a prop improperly deprecated by a sister team by digging through logs, opened a draft PR with the fix, and coordinated cross-team to unblock the customer release.

Software Engineer Intern **Mar 2024 – Sep 2024**
Technical Consulting and Research, Inc. *Weston, CT*

- Joined as the sole backend-focused intern on a small team building an internal e-commerce website from scratch, owning architecture decisions from schema design through access control.

- Owned the end-to-end backend architecture for a greenfield e-commerce platform (0→1), designing a scalable Django/Python schema to support multi-tenant inventory logic and 100+ active users.
- Designed and implemented a strict Role-Based Access Control (RBAC) system via Auth0, securing user data and ensuring compliance with enterprise security requirements.
- Built an internal audit dashboard using JavaScript and SQLAlchemy, enabling administrators to visualize authorization requests and manage security policies in real-time.
- Executed test plans for 17+ IAM roles, ensuring compliance with security requirements and preventing breaches.

Software Engineer Intern **Jun 2023 – Aug 2023**
Staples Inc. *Framingham, MA*

- Supported a critical legacy database migration by building a backend API microservice for data reconciliation and validation, including pulling and cross-checking data sourced from Salesforce.
- Built a high-throughput reconciliation engine using Java and Spring Boot to audit cross-database consistency, improving data validation reliability by 70% during a critical legacy migration.
- Optimized database performance by analyzing execution plans and implementing strategic indexing, significantly reducing query latency for reporting services.
- Engineered concurrent batch processing pipelines (Spring Batch) to handle 500,000+ records with low latency, ensuring data integrity across distributed systems.
- Automated CI/CD workflows using Jenkins, increasing development and testing efficiency.

Software Engineer Intern **Mar 2022 – Aug 2022**
PricewaterhouseCoopers *Remote, India*

- Contributed to a high-frequency financial trading system while leading identity and access management modernization across the organization, mentoring junior engineers along the way.
- Led the implementation of enterprise-wide Identity Access Management (IAM) using Okta SSO, enforcing least-privilege security policies and automating user lifecycle management.
- Co-architected a high-frequency financial system using Java, mentoring junior engineers on best practices for building robust, ACID-compliant transaction layers.

Skills

Frontend: React.js, TypeScript, JavaScript, Next.js, React Flow, Three.js
Backend: Python, Java, Kotlin, SQL, Spring Boot, Django, Node.js, Express, RESTful APIs, Batch Processing
AI & ML: LLM Agents, LangGraph, Retrieval-Augmented Generation (RAG), LLM-as-Judge Evaluation, Extend.ai
Cloud & DevOps: AWS (CDK, CloudFormation, Lambda, S3, ECR, SNS, SST), Docker, Git, CI/CD Pipelines (Jenkins)
Databases: PostgreSQL, MySQL, SQLAlchemy

Education

Southwest Baptist University **Expected Sep 2030**
Ph.D. in Artificial Intelligence *Bolivar, MO*

Boston University **Jan 2024**
Master of Science in Computer Science, Specialization in Data-Centric Computing *Boston, MA*

St. Joseph's College of Engineering **May 2022**
Bachelor of Engineering in Computer Science and Engineering *Tamil Nadu, India*